Project Proposal GIS

April 8, 2025

Project Proposal: Optimal Placement of Solar Farms in Sindh & Balochistan

Group Number 8
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User/Decision Maker

Government agencies, renewable energy planners, and private investors looking to develop solar energy projects.

Decision-Making Insights Provided by the GDSS

- Identification of high solar radiation zones suitable for solar farm installation.
- Analysis of land suitability based on terrain, elevation, and slope.
- Filtering of unsuitable areas (urban regions, water bodies, protected land) to refine site selection.
- Generation of a geospatial suitability map to assist in decision-making.

Data Requirements & Sources

Solar Radiation Data:

- MODIS MCD18A1: MODIS MCD18A1

Elevation & Terrain Data:

- NASA NASADEM HGT 001: NASA NASADEM HGT 001

Land Use & Availability:

- ESA WorldCover 10m v100: ESA WorldCover

International Boundary:

- Large Scale International Boundary: World Boundaries Data

Output:

Result:

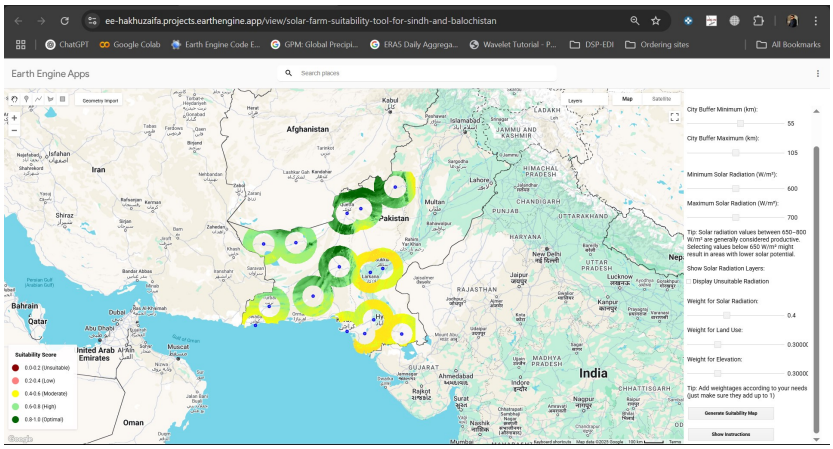

Figure 1: Final Output

Main Screen:

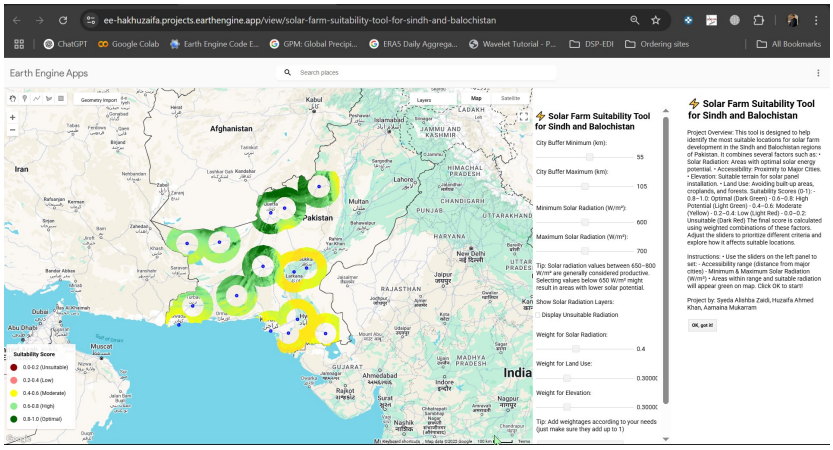

Figure 2: Deployed WebApp

Flowchart:

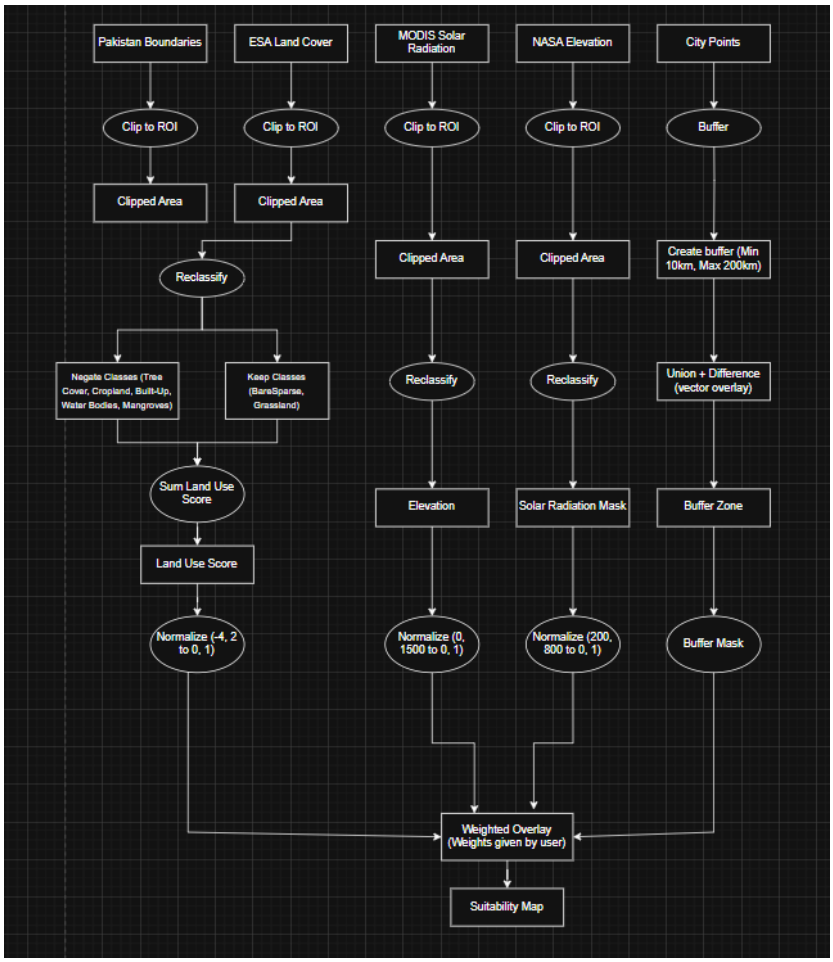

Figure 3: Flowchart of Solar Farm Suitability Tool.

Web Application UI

You can access the application here:
`Press this to redirect towards the app.`